SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA5101
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: **Total Marks: 50**

Annexure A

Code of Conduct:

I. J. Lakshmi Devi (192524300) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

J. Laksh Dev



Annexure A
Analytical Problem Solving

1. RSA Key Generation and Security Analysis
Given:
$p = 17$, $q = 19$
Compute n and $\phi(n)$
Choose $e = 5$ and calculate the private key d
Encrypt the message $M = 7$
Decrypt the ciphertext and verify correctness
Analyze the impact of choosing a small prime number on RSA security

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2. ElGamal Encryption Analysis
Given:
$p = 23$, $g = 5$, private key $x = 7$, random $k = 3$, message $M = 10$
Compute the public key
Perform encryption to generate ciphertext pair (C1, C2)
Decrypt the ciphertext to retrieve the original message
Analyze how randomness (k) improves security in ElGamal

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3. RSA Digital Signature Verification
Given:
$p = 11$, $q = 17$, $e = 7$, message $M = 8$
Compute n and $\phi(n)$
Find private key d
Generate the digital signature for the message
Verify the signature using the public key
Analyze how digital signatures ensure non-repudiation

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4. Diffie-Hellman with Attack Resistance Evaluation
Given:
$p = 29$, $g = 2$
User A private key = 5
User B private key = 12
Compute public keys for both users
Compute the shared secret key
Analyze what happens if an attacker intercepts and replaces public keys
Suggest a cryptographic solution to ensure authenticity

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5. Hybrid Encryption System Analysis
Given:
A system uses RSA for key exchange and AES for data encryption
RSA encryption time = 5 ms per key exchange
AES encryption time = 0.5 ms per data block
Total data blocks = 500
Calculate total time taken for encryption
Compare with using only RSA for encrypting all data

Analyze why hybrid encryption is more efficient
Justify its use in modern secure communication systems

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process

Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1. RSA key Generation and Security Analysis

Given:

$p = 17$

$q = 19$

$e = 5$

$H = 7$

Step 1: Calculate n

The modulus n is calculated by multiplying the two prime numbers.

$$n = p \times q$$

$$n = 17 \times 19 = 323.$$

Step 2: Calculate Euler's totient function,

Euler's totient function determine the number of positive integers less than n that are relatively prime to n.

$$\phi(n) = (p-1)(q-1)$$

$$\phi(n) = (17-1)(19-1)$$

$$\phi(n) = 16 \times 18 = 288.$$

Step 3: Generate the public and private keys.

The public exponent is given as $e = 5$ It must satisfy the condition:

$$\gcd(e, \phi(n)) = 1$$

Since gcd (5, 288) = 1, the [illegible] valid.

The private key d is the modular multiplicative inverse of e.

$$ed \equiv 1 \pmod{288}$$

$$5d \equiv 1 \pmod{288}$$

Using d = 173

$$5 \times 173 = 865 = 3(288) + 1$$

$$\boxed{d = 173}$$

Public key: $(e, n) = (5, 323)$

Private key: $(d, n) = (173, 323)$

Step 4: Encrypt the message

RSA encryption formula:

$$C = M^e \bmod n$$

$$C = 7^5 \bmod 323$$

$$7^5 = 16807$$

$$16807 \bmod 323 = 214$$

∴, the ciphertext is 214

Step 5: Decrypt the ciphertext.

RSA decryption formula:

$$M = C^d \bmod n$$

$$M = 214^{173} \bmod 323$$

Using modular exponentiation, the result is:

$M = 7$

The decrypted value matches the original message, confirming the correctness of RSA encryption and decryption.

Step 6: Security analysis of RSA

RSA is an assymetric cryptographic algorithm that uses two different keys: a public key for encryption and a private key for decryption.

Advantages:

* Provides confidentiality when used with appropriate encryption padding

* Supports digital signatures when used with suitable signature schemas.

* Allows public key to be shared without revealing the private key

* Can be used to establish or protect cryptographic keys.

Limitations:

* Small prime numbers make the modulus easy to factor.

* An attacker can calculate the private key if the Prime factors are discovered.

* Incorrect padding or weak random number generation can introduce vulnerabilities

* RSA operations can be slower than modern symmetric encryption algorithms.

Impact of small prime numbers:

In this example, $p = 17$ and $q = 19$ produce $n = 323$ which is extremely small. An attacker can easily factor 323 and recover the private key. Real-world RSA generally uses a modulus of at least 2048 bits, with secure padding and appropriate implementation.

2. ElGamal Encryption Analysis.

Given:

$p = 23 \quad g = 5 \quad x = 7 \quad k = 3 \quad M = 10$

Step 1: Calculate the Public key.

The modulus n is calculate based on the discrete logarithm problem.

The public key component is calculated using:

$$y = g^x \bmod p.$$

Substituting the given values:

$$y = 5^7 \bmod 23$$

Calculate the powers:

$$5^2 \bmod 23 = 2$$

$$5^4 \bmod 23 = 4.$$

Therefore,

$$5^7 \bmod 23 = (5^4 \times 5^2 \times 5) \bmod 23$$

$$y = (4 \times 2 \times 5) \bmod 23$$

$$y = 17.$$

Public key: $(p, g, y) = (23, 5, 17)$

Private key: $x = 7.$

Step 2: Generate the ciphertext pair

ElGamal encryption produces two ciphertext values C_1 and C_2.

The formulas are:

$$C_1 = g^k \bmod p.$$

$$C_2 = M \times y^k \bmod p.$$

Calculate C_1

$$C_1 = 5^3 \bmod 23$$

$$C_1 = 125 \bmod 23$$

$$C_1 = 10$$

Calculate C_2

First, calculate y^k.

$$y^k = 17^3 \bmod 23$$

Since $17^2 \bmod 23 = 13$

$17^3 \bmod 23 = 13 \times 17 \bmod 23 = 14.$

Now Calculate C_2:

$$C_2 = 10 \times 14 \bmod 23$$

$$C_2 = 140 \bmod 23$$

$$C_2 = 2.$$

Therefore, the ciphertext pair is:

$$(C_1, C_2) = (10, 2)$$

Step 3: Decrypt the cipher text

First, calculate the shared secret:

$$S = C_1^x \bmod p.$$

$$S = 10^7 \bmod 23.$$

Using modular exponentiation:

$$10^2 \bmod 23 = 8$$

$$10^4 \bmod 23 = 18$$

$$10^7 \bmod 23 = (10^4 \times 10^2 \times 10) \bmod 23$$

$$S = (18 \times 8 \times 10) \bmod 23$$

$$S = 14.$$

Next, calculate the modular inverse of 14 modulo 23

$$14^{-1} \bmod 23 = 5$$

Verification:

$$14 \times 5 = 70 \equiv 1 \pmod{23}$$

Now recovers the original message:

$$M = C_2 \times S^{-1} \bmod p.$$

$$M = 2 \times 5 \bmod 23.$$

$$M = 10.$$

The original Message is successfully recovered.

Step 1: Analyse the role of randomness k.

The random value k is an important part of ElGamal encryption. It is used to generate the cipher text for each encryption operation.

Importance of k:

1. Ciphertext variation: Encryption the same message with different valid random value normally produces different ciphertexts.

2. Confidentiality: An unpredictable k makes it harder for attackers to determine the shared secret.

3. Protection against repeated patterns: Fresh randomness reduces the risk of exposing information through repeated ciphertexts.

4. Secure implementation: The value of k must be generated securely and should not be reused carelessly.

5. Parameter requirements: The random value must follow the algorithm's valid range and group requirements.

Impact of weak randomness: If an attacker can predict k then attacker may be able to calculate the shared secret and recover the message. Reusing k can also reveal relationship between messages and compromise confidentiality.

Application:

RSA:

* Digital signature and signature verification
* Key establishment and protection of cryptographic keys.
* Public-key cryptography in security protocols.

ElGamal:

* Public-key encryption in suitable cryptographic systems.
* Cryptographic protocols requiring randomized encryption.
* Certain privacy-preserving and electronic voting systems